

PROBLEM STATEMENT

Quadruple Composite Chain Rule

Find the derivative with respect to x of the function:

$$y = \tan^4(\sec^2 x)$$

STEP 1 প্রদত্ত ৪-স্তরীয় যৌগিক ত্রিকোণমিতিক ফাংশনটি বন্ধনী আকারে লিখি (Expressing in Explicit Power Form)

$$y = [\tan(\sec^2 x)]^4$$

STEP 2 উভয়পক্ষে x -এর সাপেক্ষে অন্তরীকরণ অপারেটর প্রয়োগ করি (Applying Differentiation Operator w.r.t. x)

$$\frac{dy}{dx} = \frac{d}{dx} ([\tan(\sec^2 x)]^4)$$

STEP 3 সর্ববহিঃস্থ ঘাত ৪-এর জন্য শক্তি বিধি (Power Rule) প্রয়োগ করি (Applying Power Rule to Outer Exponent)

$$\frac{dy}{dx} = 4 [\tan(\sec^2 x)]^{4-1} \cdot \frac{d}{dx} [\tan(\sec^2 x)] = 4 \tan^3(\sec^2 x) \cdot \frac{d}{dx} [\tan(\sec^2 x)]$$

STEP 4 ট্যানজেন্ট ফাংশনের জন্য অন্তরজ নির্ণয় করি (Differentiating Tangent Function via Chain Rule)

$$\frac{d}{dx} [\tan(\sec^2 x)] = \sec^2(\sec^2 x) \cdot \frac{d}{dx} (\sec^2 x)$$

STEP 5 সর্বঅভ্যন্তরীণ $\sec^2 x$ -এর অন্তরজ নির্ণয় করে সামগ্রিক গুণফল সাজাই (Differentiating Innermost Secant-Squared & Combining Factors)

$$\frac{d}{dx} (\sec^2 x) = 2 \sec x \cdot \frac{d}{dx} (\sec x) = 2 \sec x (\sec x \tan x) = 2 \sec^2 x \tan x$$

$$\frac{dy}{dx} = 4 \tan^3(\sec^2 x) \cdot \sec^2(\sec^2 x) \cdot [2 \sec^2 x \tan x] = 8 \sec^2 x \tan x \tan^3(\sec^2 x) \sec^2(\sec^2 x)$$

FINAL ANSWER
$$\frac{dy}{dx} = 8 \sec^2 x \tan x \tan^3(\sec^2 x) \sec^2(\sec^2 x)$$

BOX 1: MATHEMATICAL FOUNDATIONS & FORMULA REFERENCE

- 4-Stage Chain Rule Composition:** Let $u = \tan v$, $v = w^2$, and $w = \sec x$. Then $y = u^4$.
$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dw} \cdot \frac{dw}{dx} = (4u^3) \cdot (\sec^2 v) \cdot (2w) \cdot (\sec x \tan x)$$
- Secant Derivative:** $\frac{d}{dx}[\sec x] = \sec x \tan x$.
- Secant-Squared Derivative:** $\frac{d}{dx}[\sec^2 x] = 2 \sec^2 x \tan x$.

BOX 2: COMMON STUDENT TRAPS & PITFALLS

- Overlooking the Inner Power of Secant:** Differentiating $\sec^2 x$ simply as $\sec x \tan x$ without the outer factor of 2.
- Argument Nesting Confusion:** $\sec^2(\sec^2 x)$ is the square of secant evaluated at angle $\sec^2 x$. It is not $\sec^4 x$.
- Constant Multiplier:** The coefficient becomes $4 \times 2 = 8$. Missing one of the power multipliers leads to 4 or 2 instead of 8.

BOX 3: REAL DOMAIN & DIFFERENTIABILITY ANALYSIS

- Base Secant Singularity:** $\cos x \neq 0 \implies x \neq \frac{\pi}{2} + k\pi$ for $k \in \mathbb{Z}$.
- Outer Tangent Singularity:** $\sec^2 x \neq \frac{\pi}{2} + m\pi$. Since $\sec^2 x \geq 1$, this excludes specific points where $\sec^2 x = \frac{\pi}{2} + m\pi$ ($m \geq 0$).
- Differentiability:** Smooth and differentiable on all intervals avoiding these singular points.

BOX 4: EXPLICIT VARIABLE SUBSTITUTION HIERARCHY

- Let $u = \sec x \implies u' = \sec x \tan x$
- Let $v = u^2 = \sec^2 x \implies v' = 2uu' = 2 \sec^2 x \tan x$
- Let $w = \tan v = \tan(\sec^2 x) \implies w' = \sec^2 v \cdot v' = 2 \sec^2 x \tan x \sec^2(\sec^2 x)$
- Finally $y = w^4 \implies y' = 4w^3 w' = 8 \sec^2 x \tan x \tan^3(\sec^2 x) \sec^2(\sec^2 x)$

BOX 5: EXAM MARKS BREAKDOWN (AMIE SECTION-A FORMAT)

- Step 1–2 (4-Tier Chain Rule Formulation):** 1.0 Mark
- Step 3 (Outer Power-4 Differentiation):** 1.0 Mark
- Step 4 (Tangent Derivative):** 1.0 Mark
- Step 5 (Inner $\sec^2 x$ Derivative & Final Form):** 1.0 Mark
- Total Allocation:** 4.0 Marks